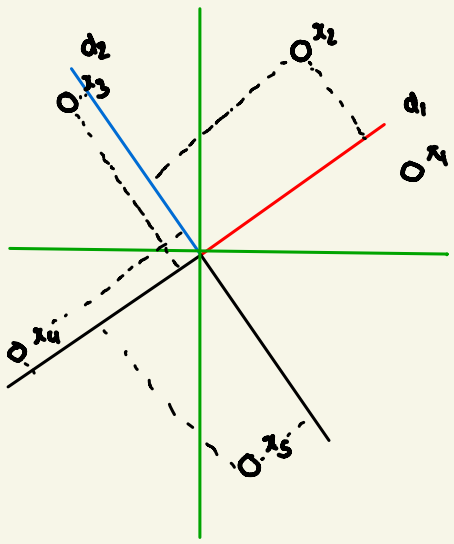
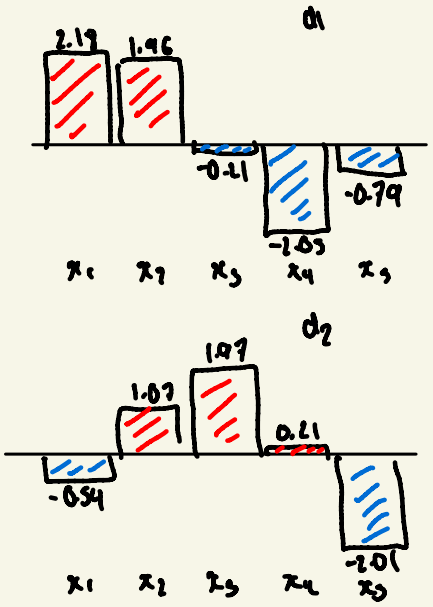
①





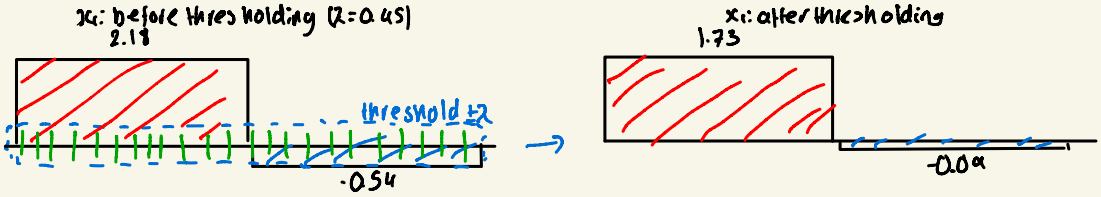
②

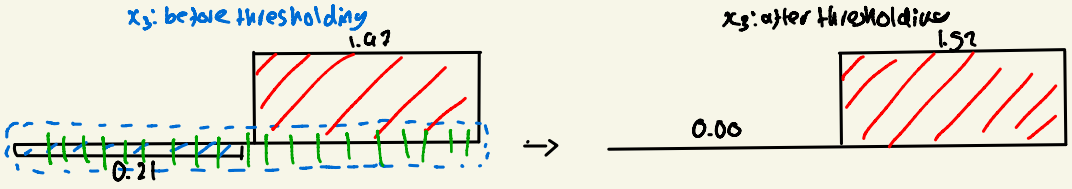
$$\text{minimize } ||x - a_1 d_1 - a_2 d_2||^2 + \lambda(|a_1| + |a_2|)$$

③

Dense coefficient matrix A (no penalty)

	a_1	a_2
x_1	2.18	-0.54
x_2	1.96	1.07
x_3	-0.21	1.97
x_4	-2.05	0.21
x_5	-0.79	-2.01





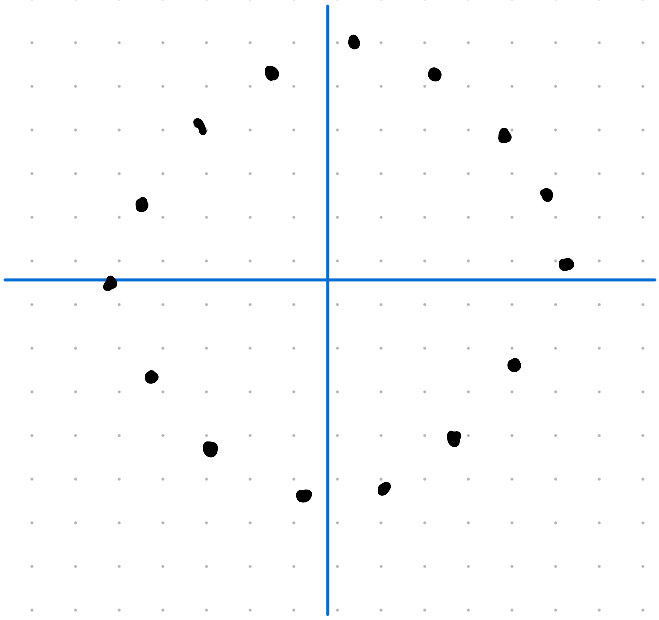
Dense A

	a_1	a_2
x_1	2.1x	-0.54
x_2	1.46	1.07
x_3	-0.21	1.47
x_4	-2.05	0.21
x_5	-0.79	-2.01

Sparse A (λ=0.45)

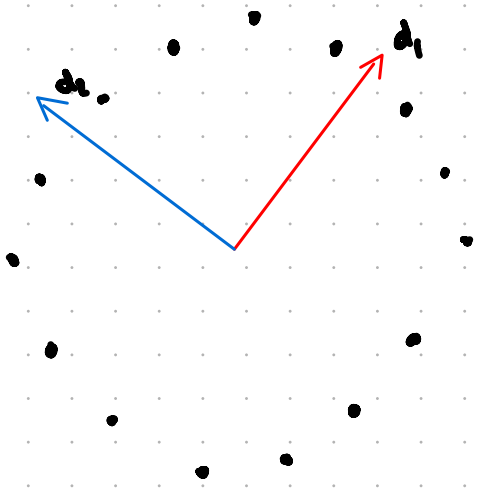
	a_1	a_2
x_1	1.73	-0.09
x_2	1.51	0.02
x_3	0.00	1.52
x_4	-1.60	0.03
x_5	-0.34	-1.56

④



Start with D from SVD
Fix D → find sparse A (Lasso)
Fix A → update D (SVD of residuals)
Go to step 2

⑤



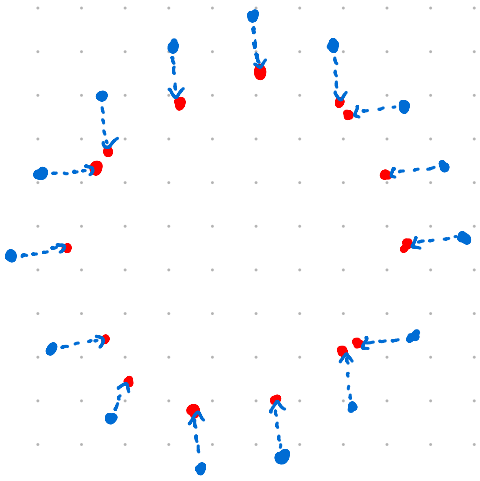
Iteration 0: SVD initialisation

$d_1 = [0.61, 0.80]$
$d_2 = [-0.90, 0.61]$

These are principle axes of the data eclipse

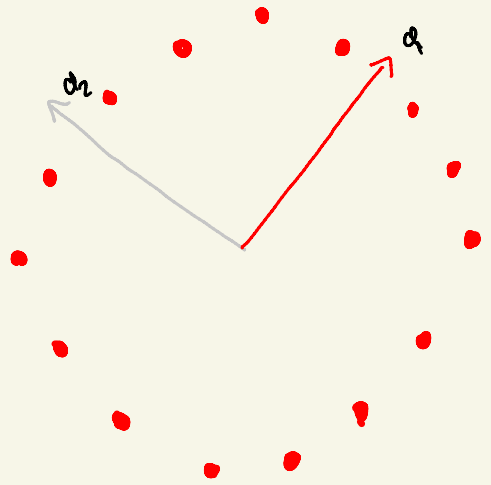
Best for dense reconstruction but we want sparse

⑥



After sparse coding
Total error: 0.3

7



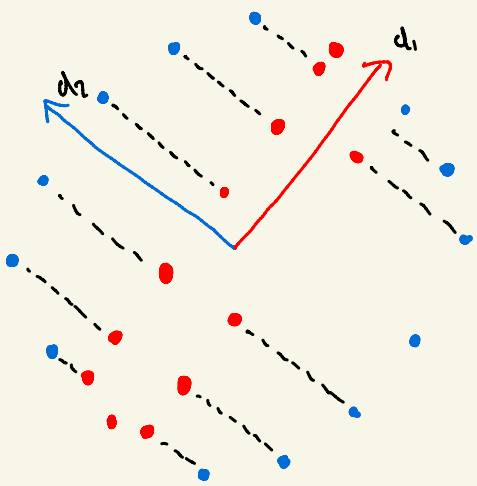
Updating d_1 (K-SVD method)

Collect all points that USE d_1 ($a_1 \neq 0$).

Found 14 points that use d_1

Points where $a_1 = 0$ don't care about d_1 at all, so we ignore them

8

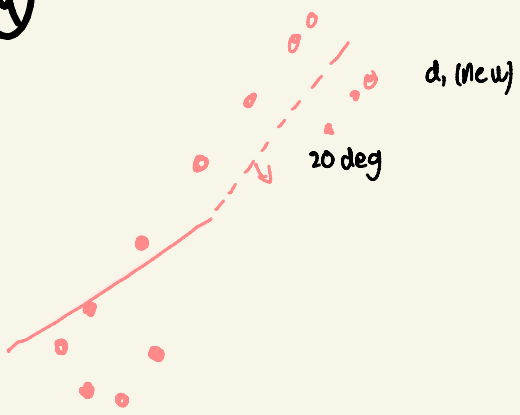


Remove d_2 contribution

For each user a_1 compute residual: $x - a_2 \cdot d_2$

This strips away everything that d_2 explains, leaving only what d_1 is responsible

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Rank-1 SVD of residuals

Take the residual cloud (red) and find its principal direction using SVD – just like step 1

The first singular vector of these residuals become the new d_1

Dashed = old d_1
Solid = new d_1
Rotated 20 degrees

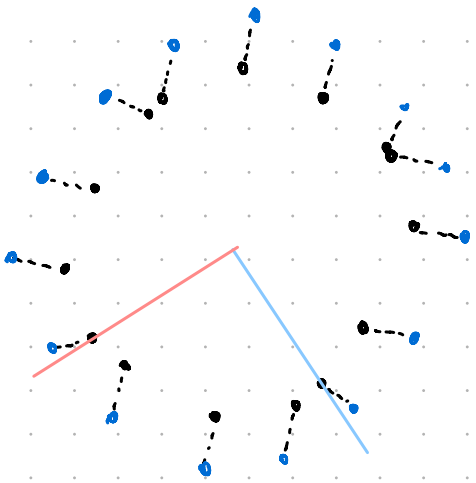
(d)

Repeat for d_2

Same process:
Collect d_2 users
Remove d_1's contribution
SVD of residuals → new d_2

Now both atoms have moved
Dashed = old pos
Solid = new pos

(i)



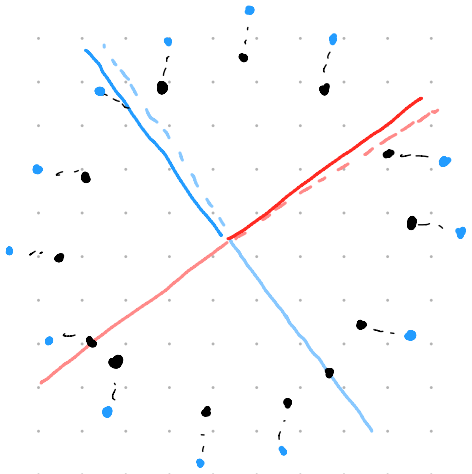
Iteration 1
Total error: 7.7

re-run Lasso
with UPDATED atoms

The sparse codes change too,
because the atoms moved

Different atoms → different
dot products → different ?

(ii)

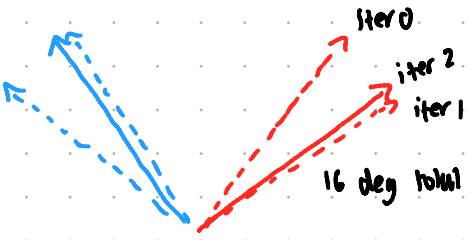


Iteration 2
Total error: 7.0

updated
using residual from iteration 1

Atoms shift more
Each iteration they move
less as the converge

13



full trajectory of d1:

Started: [0.61, 0.79]
After 1: [0.84, 0.54]
Final: [0.90, 0.60]

Each iteration, the atom rotates a bit toward to reduce the sparse le construction error

SVD atoms (iter 0)
Total error: 8.5

Learned atoms
Total error 7.0

16% less error

D is not fixed

SVD → initial D
Lasso → Sparse A
SVD of residuals → better D
Repeat